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Deep learning paradigms in remote sensing with a comparative evaluation of CNNs and transformers for satellite imagery

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Abstract

Satellite Image Classification is a process in remote sensing that aims at classifying pixels of satellite imagery into usable information of land use and land cover. This information is not only limited to several types of land covers and specific features, but it is an essential tool for pattern recognition and storing land information. Traditionally, convolutional neural networks (CNNs) have been popularly used in this domain due to their feature-extraction capabilities on grid-like data. Recent advances in the transformer model have also introduced a new way of performing tasks and working with the labelled data. This study compares six CNNs models (CNN, InceptionV3, VGG16, Xception, ResNet50, ResNet101) and a Transformer model (Google ViT-Base-Patch16-224-in21k) on the UC Merced Land Use dataset. All models were trained and evaluated under identical computational conditions using a single NVIDIA A100 GPU with fixed hyperparameters and standardized pre-processing of the same dataset. The objective was to determine the effectiveness of architectures with a trade-off between accuracy and computational efficiency. Evaluation runs showed that ResNet101 (96.33% accuracy with 1m18sec of training time) demonstrated a competitive performance in producing results similar to Google ViT (98.57% accuracy with 9 m 28 s of training time) while taking less than 1/7th of the Transformer's training time.

Keywords Deep learning (DL), Transformer model, Google ViT, Convolutional neural networks (CNN), Satellite image classification (SIC)

1 Introduction

Multi-class classification is an important task in remote sensing, where images may contain one of several attributes or labels, particularly about land cover or objects that must be labelled. This classification enables analysis of patterns in the environment and helps in tasks such as urban, forest planning and disaster management by providing detailed and relevant information. Availability of high-quality satellite images with high resolution, diverse landforms, and multi-class classification is a challenging task, which requires robust and advanced classification techniques. Satellite image classification



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is an important technique to analyze and extract valuable information regarding land use, land cover, object density, and other surface characteristics of an area [1]. A satellite image classification task enables highly advanced land cover analysis, benefiting multiple sectors such as urban planning, agriculture and disaster management. Traditional computer vision techniques lack robust feature extraction capabilities, but instead, use algorithms whose implementation may require expertise and time. Due to this, deep learning (DL) techniques are preferable for classification tasks for remote sensing images [2]. In comparison, DL techniques are far more efficient and flexible and offer ease of use through automation. A complex and large dataset of low-quality images can be processed automatically via DL without the need for excessive pre-processing on the raw data [3]. The satellite imagery contains diverse data with variations in resolution, land cover, and seasonal changes, so DL has proven to be far more efficient in handling and learning the underlying features of massive data than traditional computer vision techniques [4][5]. The limitations to the use of DL methods (such as the lack of labelled data and the need for expensive computational resources) can be mitigated by the intelligent use of DL techniques[6] [7].

The use of robust and highly advanced deep learning models, such as transformers can lead to advancements in this field by improving the accuracy, efficiency, and scalability of satellite image classification and analysis [8, 9]. The Transformer model's ability to adapt to huge datasets and architectures with the help of attention mechanisms can help address the key challenge [10]. Multiple challenges arise in multi-class classification due to the lack of high-resolution data, mixed pixel objects, and various landscapes [11]. Additionally, the lack of annotated datasets, computational power, and expertise required to fine-tune the algorithms also adds to the challenges [3, 4]. Pixel-wise classification identifies each pixel class in remote sensing images, hyperspectral images (HSI), and high-resolution satellite images, automatically assigning a label to the satellite images [12]. For data scarcity in satellite imagery, a semi-supervised approach was proposed to mitigate the problem of fully supervised deep networks requiring manual annotation of the dataset [13][14]. The approach used in this study involved using a general semi-supervised framework (GSF) that requires a small amount of an annotated dataset and proposed using a pre-trained CNN model for drawing out global-level features of remote sensing images (HRRS) in one of the tasks. This approach significantly improved accuracy by combining XGBoost-based classifiers, which increase accuracy through iterative computation of weak classifiers [15]. CNN has difficulty capturing contextual relationships between multiple objects, so to solve this problem, an enhanced Multi-layer Perceptron (MLP) driven by an Adagrad-based optimizer was used as the classifier [11]. Another study [16] proposed a CNN-MLP model using augmented data, which was evaluated on three public remote sensing datasets (UC-Merced Land Use, AID, and NWPU-RESISC45). The proposed method achieved better classification performance than current state-of-the-art classification techniques, highlighting the potential of deep network architectures in extracting contextual understanding within an image [2, 17]. While DL models have been compared in image classification of remote sensing imagery, there is a lack of systematic comparison between transformer-based and traditional convolution-based models and architectures. Analysis of 42 experiments in the past study [18] yielded valuable insights into transformer-based fine-tuning applications and helped gain a better understanding of image classification.

Studies in the past [8, 19] have utilized transfer learning for land use and land cover models, which have proven to be effective, with both Google's ViT base and ViT large properly adjusted, leading to the Captum powered interpretable vision transformer for land use and land cover classification. It provides insight into monitoring biases while offering transparent, interpretable solutions with current-generation transformer models. The use of interpretability with Captum does not require significant changes to the model architecture and provides comprehensive information about the parts of the image and model layers that most impact the final prediction. It also presents the use of Swin Transformers (SwinT) for land use and land cover classification, which showed a slight drop in accuracy with increased efficiency in training. Training DL models through labelled data is a significant challenge in the remote sensing domain. Researchers in the past have conducted studies that have used current-generation transformer models to address the above challenge. To address the need for large amounts of labelled data, one study has proposed pre-training a visual transformer with unlabelled remote sensing images [7, 17]. The study used a transformer pre-trained on unlabelled images and then refined the model on satellite image time series (SITS). The results showed better predictions on datasets with similar labels. This was possible due to the increased familiarity with the provided images of the model. The technique of Generative Adversarial Networks (GAN), to create images based on the statistical distribution of images captured by satellites, thereby overcoming the lack of data produced. GANs were used to generate images based on the data, then Vision Transformers (ViT) were trained to classify satellite imagery. The results showed that when data augmentation was applied to the original dataset and a GAN was used to create more data points, the accuracy of ViT increased from 76.9% to 98.7%. The approach used is a complex and lengthy process that involves careful training of the GAN model to avoid biases during training of ViT on the resulting dataset [8].

The pattern of images in remote sensing and the classification of remote sensing images has always been challenging. Based on the challenges in the previous studies, we were motivated to conduct the current study. The current study revealed that traditional models were not suitable for extracting deep spectral features from multispectral images. Hence, an adaptation was made to CNN models. The study also emphasized on challenges such as high dimensionality, limited labelled samples and complex spatial structures arising in satellite data. With such challenges, the study highlighted the need for more specialized architectures for high-dimensional data, which mitigate the unfairness and lack of annotations across different datasets. The main objective of this research was to provide a systematic analysis of deep learning architectures for remote sensing data in real-world applications. This approach has led to more efficient and scalable deployment of these architectures. This was done by bridging the gap in accurate information on model performance, especially in classifying complex landforms and managing diverse terrain features. Consequently, the need for timely and effective policy formulation and resource allocation in various land monitoring and management areas can be met.

This research paper presents a structured comparison of the effectiveness and capability of CNN (CNN, InceptionV3, VGG16, Xception, ResNet50, ResNet101) and Transformer (Google ViT) models in multi-class classification of satellite images. The models were evaluated, and their performance compared on the UC Merced Land Use dataset,

with findings the potential of transformer model to increase accuracy and scalability in satellite image classification.

The rest of the paper is divided into the following sections: Sect. 2 details the datasets and pre-processing, Sect. 3 describes the methodologies and their structure, and Sect. 4 discusses the results, analysis, and conclusions added in Sect. 5. The declaration has been added in Sect. 6.

2 Dataset and pre-processing

The UC Merced land use dataset (<http://weegee.vision.ucmerced.edu/datasets/landuse.html>) is a widely used benchmark dataset for remote sensing and computer vision. It comprises aerial images captured by the United States Geological Survey (USGS) National Map Program over the Merced County area in California. The dataset consists of 2100 images, with 100 images from each of the 21 distinct classes representing different land use and land cover categories, including agricultural, airplane, baseball_diamond, beach, buildings, chaparral, dense_residential, forest, freeway, golf-course, harbour, intersection, medium-residential, mobile-home-park, overpass, parking-lot, river, runway, sparse-residential, storage-tanks and tennis-court as shown in Fig. 1. Each image measures 256×256 pixels. Since there is no disparity in the class distribution and the dataset is balanced, this dataset is used for evaluating models in the satellite image classification techniques. For preparing the dataset, each image has simply been converted to tensors of size (256, 256, 3) and normalised into a uniformly scaled dataset. The pixel intensity distribution analysis of the dataset shows a balanced distribution of pixel intensities, indicating consistent brightness and contrast levels across the images. The images are well-suited for classification tasks without the need for extensive data augmentation and pre-processing.

3 Methodology for comparing deep learning techniques

This section details the architecture flow of the models and outlines the reasons for choosing them. Our study aims to identify model that is effective for satellite image classification. This was done via comparison of seven models which are: CNN, InceptionV3, Xception, VGG16, ResNet50, ResNet101, and Google Vision Transformer (ViT). The

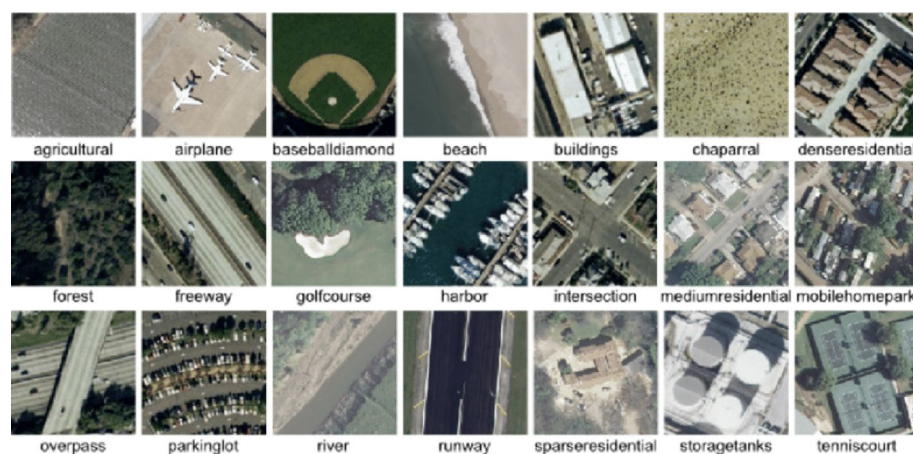


Fig. 1 The 21 Classes of UC merced land use dataset

steps involved in the methodology for satellite image classification by comparison of various DL techniques is listed below:

Step 1 Collect the labelled dataset of satellite images. This was done from the UC Merced site.

Step 2 Data visualization and analysis for checking the noisy data.

Step 3 Data pre-processing and augmentation to resize, augment, and normalize the dataset.

Step 4 Splitting the dataset for training that involves conversion of images to tensors and split the dataset into train, test, and validation sets.

Step 5 Training of the models includes six CNN models and a Transformer model on the same dataset.

Step 6 Evaluating the accuracy, precision, F1score, recall, etc.

Step 7 Comparing the model performance to determine the best-performing model.

3.1 Convolution neural network (Traditional)

Convolutional neural network is a specialised neural network designed for processing structured grid data like images [15]. The Layers of CNN are organised in 3 dimensions length, width, and colour. It is a feed-forward network consisting of multiple filters which glide over the input image and encapsulate data within a window to a single data point in turn preserving patterns within a grid. The convolutional layer is the main building block of a CNN containing a set of filters typically followed by an activation function and a pooling layer for reducing the overall computational complexity of a step by reducing dimensionality [20]. The final layers made up of fully connected layers (multilayer perceptron) perform the classification by integrating the extracted features. During the training, the neural network optimises filters and weights to minimise the difference between its predicted label and the actual labels, giving an effective model for image classification tasks as shown in Fig. 2.

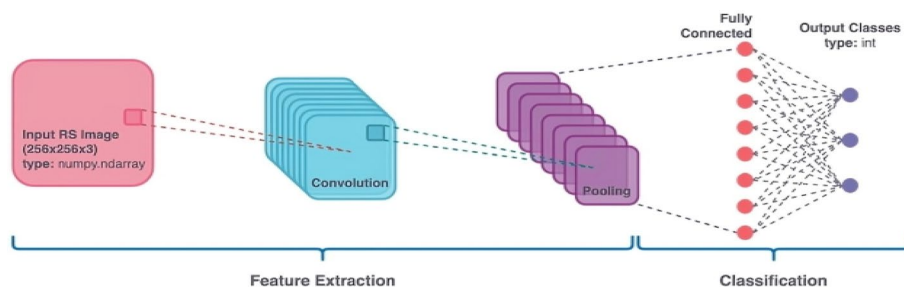


Fig. 2 Architecture of traditional convolutional neural network model

3.2 InceptionV3

It is a pre-trained deep CNN architecture developed by Google in 2015, that builds upon the traditional CNN architecture with the concepts of the 3 inception modules consisting of a convolution, pooling, and fully connected layer for image classification tasks [17, 21]. It enhances accuracy and efficiency while using lesser computational power. The network can perform convolutions with multiple filter sizes simultaneously thus capturing a diverse set of features at each layer using the inception modules. The design was introduced to allow for deeper neural networks by keeping the number of parameters low. The architecture in Fig. 3 enables the use of factorised convolutions or splitting of larger kernels into smaller more efficient ones and uses auxiliary classifiers which are linear SoftMax classifiers to counter the vanishing gradient problem from its previous iterations, however, differ in using 7×7 factorised convolutions and batch normalisation in its auxiliary classifiers along with RMSProp for its optimiser. Pre-trained on ImageNet, Inception V3's capabilities and multi-scale processing allow for efficient and powerful image classification, best in use for people recognition and object detection [4].

3.3 Xception

"Extreme Inception" or Xception is also a deep CNN architecture that takes inspiration from the Inception model [22]. It replaces the Inception modules for depth-wise separable convolutions. Taking an approach that entirely decouples the task of spatial correlation and cross-channel. The architecture as seen in Fig. 4, performs convolutions in a single step. It applies a depth-wise convolution to filter each input channel, separately. These are put into pointwise convolutions to combine outputs [23]. This separation allows the model to divide the process into two distinctive parts: for depth-wise convolution, a single filter is applied to each channel separately. For point-wise convolution, a 1×1 filter combines outputs of deputised convolution into a single feature map. The Exception model does this to capture spatial features and channel-wise features independently, which in turn leads to more efficient learning with fewer parameters.

3.4 VGG16

It is a pre-trained deep CNN architecture proposed by Oxford's Visual Geometry Group in 2014, designed for image classification. It performs well on image recognition and localisation tasks [11]. VGG16 is 16 layers deep and is made up of 13 convolutional

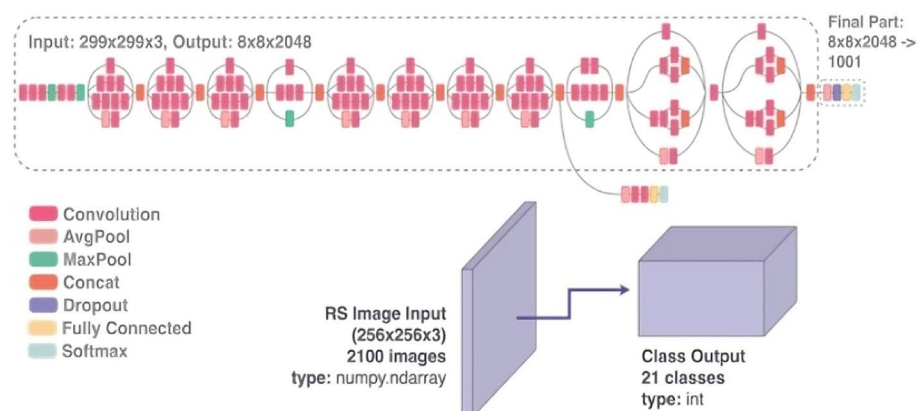


Fig. 3 Architecture of Inception V3 base model

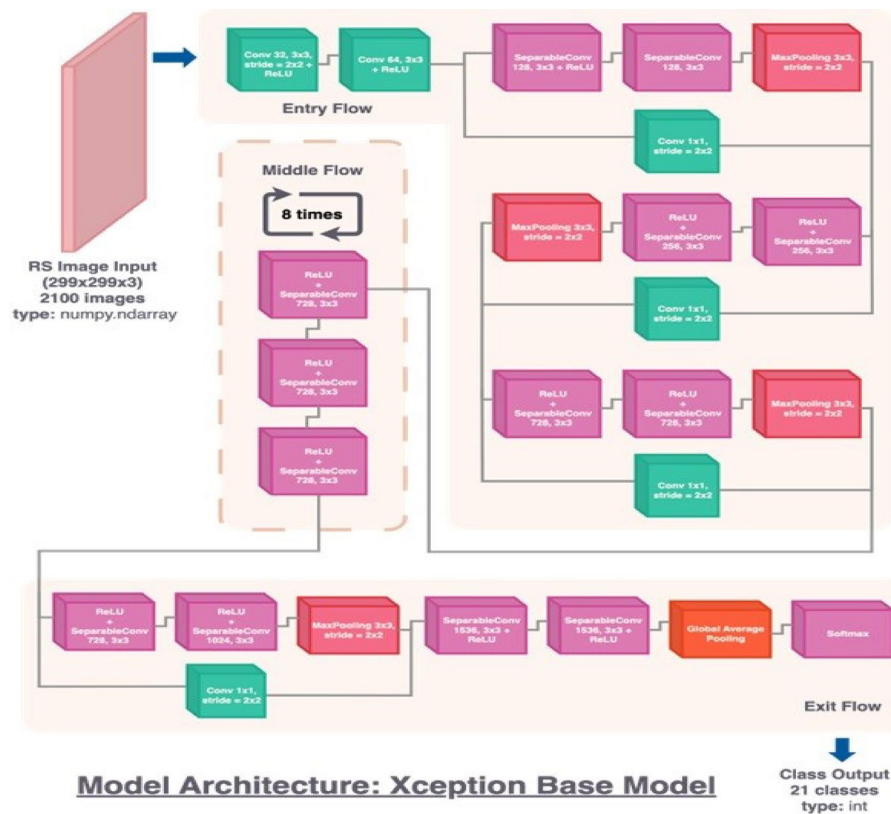


Fig. 4 Architecture of Xception base model

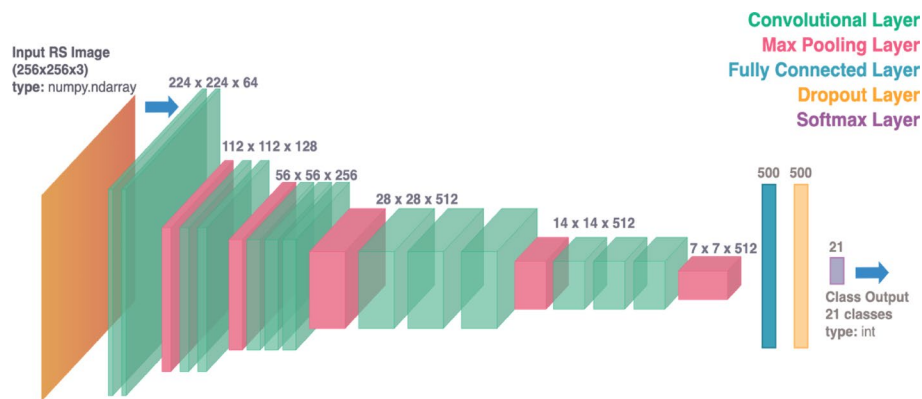


Fig. 5 Architecture of VGG16 base model

layers followed by 3 fully connected layers, as illustrated in Fig. 5. The network utilises 3×3 convolution filters throughout with ReLU activation. This is a very straightforward model architecture with convolutional layers and max pooling which reduces spatial dimensions and computational load. The important features are extracted and fed into fully connected layers to perform classification. VGG16 is pre-trained on huge datasets like ImageNet, allowing it to generalise well to various image recognition tasks by leveraging the learned representations from extensive, diverse training data.

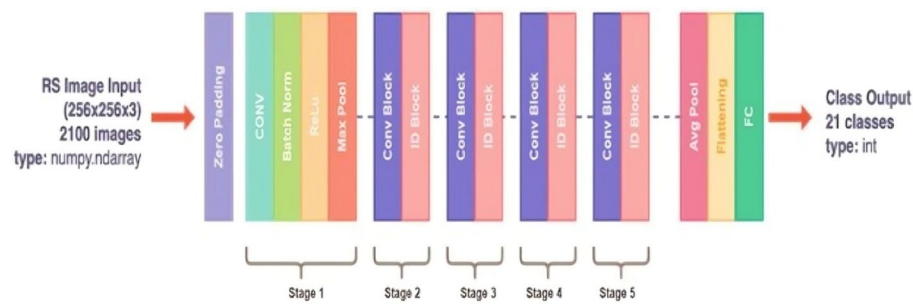


Fig. 6 Model Architecture - ResNet50 Base Model

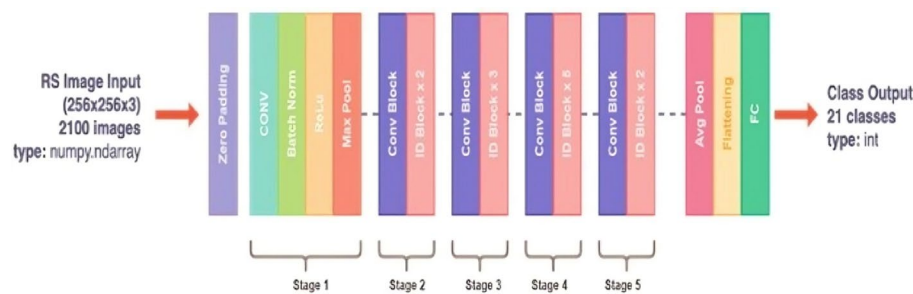


Fig. 7 Model Architecture - ResNet101 Base Model

3.5 ResNet50

Residual Network with 50 layers is a DCNN designed by Microsoft in 2015, to address the degradation problem in very deep networks [12]. The degradation problem refers to the worse performance seen when making deeper layers due to a vanishing gradient. Trained on ImageNet with 50 weight layers, ResNet50 introduces residual learning which allows for shortcut connections that skip over tricky layers, making it more efficient. These residual blocks called bottleneck residual blocks shortcut connections. They permit the network to learn residual functions instead of learning directly from the underlying mapping. This enables the network to learn identity mappings which help preserve the gradient flow. The ResNet50 architecture (Fig. 6) incorporates various residual blocks, containing three bottleneck convolutional layers with batch normalisation and ReLU activation functions. This allows the network to bypass the problem of vanishing gradient and leads to improvement of accuracy.

3.6 ResNet101

Residual Network with 101 layers builds the ResNet architecture to greater depths to capture more complex features while mitigating the degradation problem of vanishing gradient that usually occurs with deeper networks [12]. Like ResNet50, as shown in Fig. 7, it leverages residual learning, enabling the network to learn identity mappings which help preserve the gradient flow [21]. This approach aids to preserve the gradient flow during back-propagation and hence, enables the training of very deep networks. This also has identity shortcut connections that add the input of the block directly to the corresponding output. The increased depth allows ResNet 101 to capture more detailed and abstract features leading to improved performance on complex image classification tasks.

3.7 Google vision transformer (ViT-base-patch16-224-in21k)

It is a variant of the pre-trained vision transformer Google ViT, which is built upon a transformer-based deep learning architecture, see Fig. 8 for reference, originally designed for NLP tasks [19]. It can also be used for image classification. Unlike CNNs that use convolutions to extract features, ViT divides an image into a sequence of fixed size 16×16 patches, which get linearly embedded by the patch embedding layer. These embeddings are bound with positional encodings to maintain spatial information. The model processes these patches through layers of self-attention and feed-forward networks. This self-attention mechanism in ViT captures dependencies and relationships between different parts of the image more effectively than CNNs. The BERT-like architecture leverages the power of ImageNet-21 K pre-training and enhances its performance when fine-tuned on specific image classification tasks. This model has shown competitive performance compared to ResNet-based architectures when dealing with large-scale datasets [18]. The model's strong feature extraction capabilities that were demonstrated factor in for a good overview of its performance [7]. Apart from these investigates the inherent struggle of the vision-transformer model in small-scale datasets due to a lack of inductive biases [10].

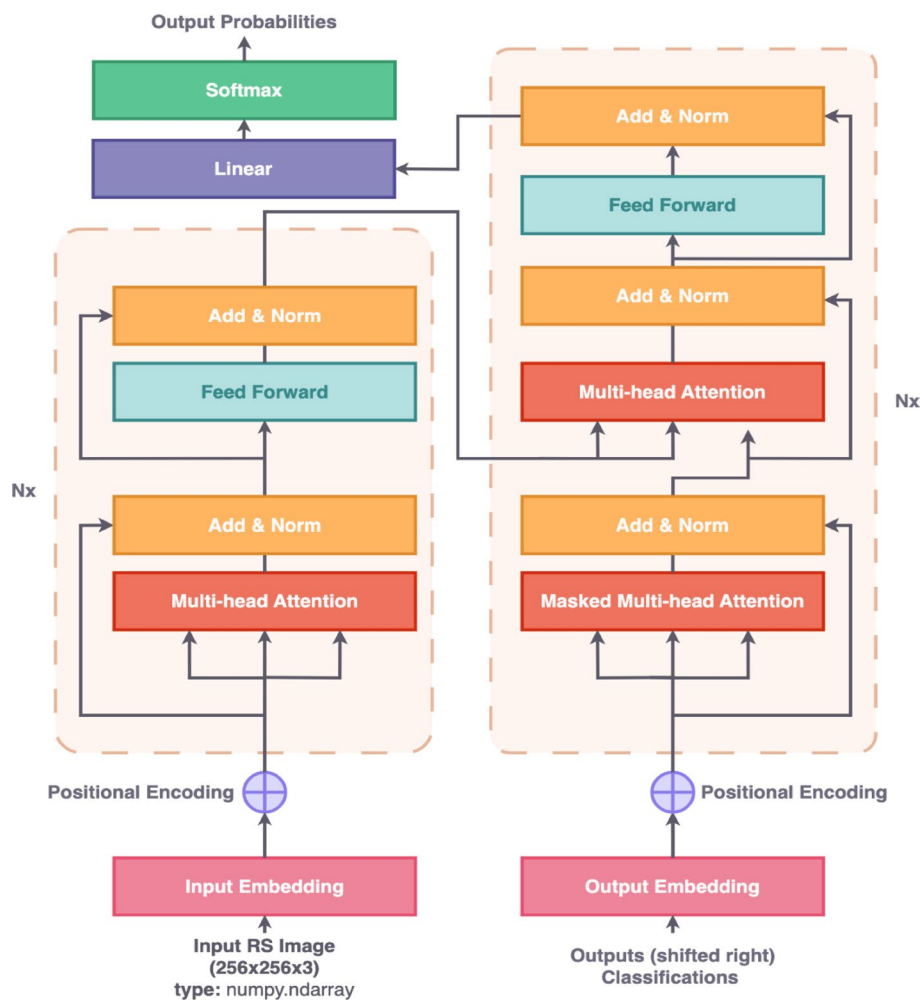


Fig. 8 Model Architecture - Google Vision Transformer

4 Result and analysis

The study was conducted on a single NVIDIA A100 GPU with estimated usage on the machine of 60 compute units where 1GB per hour = 1 compute unit. Python was the programming language and the frameworks included PyTorch 2.0 and Tensorflow 2.12. CUDA 11.8 was used for GPU support for training of the transformer model. The experiments included plotting, visualisation, and calculation helpers like Matplotlib, Numpy, Tensorboard and Weights and Biases (wandb). The training of the transformers model was aided by the HuggingFace (hf) trainer. The training and testing split ratio of the dataset was 80 and 20%.

In Table 1, ResNet101 demonstrates the highest accuracy (96.33%) amongst the CNN models, followed by ResNet50 (92.86%). This can be attributed to the deeper architectures and residual learning framework of the ResNet models compared to the CNN baseline model which could only reach 73.10% accuracy. In terms of capturing spatial details, Table 1 also highlights that InceptionV3 and Xception have shown relatively lower accuracy (InceptionV3: 54.76% and Xception: 53.57%), indicating a lagging while capturing spatial details compared to the ResNet models. Additionally, VGG16 also managed to show respectable accuracy (83.10%) crediting to the deep sequential structure for capturing hierarchical features. Comparing loss, ResNet101 again outperforms the CNN models and achieves the lowest loss of 0.2061. This loss number signifies the ResNet101 model's ability to generalize with minimum overfitting. Following closely, ResNet50 also shows comparable loss results (0.2400) to demonstrate a good convergence within the environment. In contrast, InceptionV3 and Xception faced an initial challenge to learn robust features which is also demonstrated in the higher loss values of 2.6475 for InceptionV3 and 2.4954 for Xception. VGG16's moderate loss values of 1.4025 and its downward trend in Table 2 are suggestive of moderate overfitting.

In terms of Precision, higher values for ResNet models (ResNet101: 96.72% and ResNet50: 93.39%) indicate that these models effectively reduce the number of false positive predictions compared to many shallow models like VGG16: 84.77%, CNN Baseline: 76.16%, InceptionV3: 56.52% and Xception: 55.10%. Demonstrating good precision and recall values, ResNet models have come out to be the most balanced among all the CNN models with F1-Scores of 98.65% for ResNet101 and 96.26% for ResNet50. Lower F-1 scores of InceptionV3 (55.01%) and Xception (53.71%) highlight that these models face difficulties in classification tasks for the dataset used in this study.

Demonstrating superior spatial consistency, ResNet models again achieve higher scores (ResNet101: 13.4807 and ResNet50: 13.3521) along with VGG16 (13.0543). Comparably, the baseline CNN model, InceptionV3 and Xception do not have much lower dice coefficient values (CNN: 12.6490, InceptionV3: 11.6281 and Xception: 11.9877), suggesting that these models faced an initial struggle with fine-grained spatial recognition.

Table 1 Evaluation outcomes of all the models involved in this study

Models	Accuracy	Loss	Precision	Recall	F1-Score:	Dice coefficient	Total Time
CNN	0.7310	0.8724	0.7616	0.7310	0.7378	12.649	1m26s
Inceptionv3	0.5476	53.3309	0.5652	0.5476	0.5501	11.9877	2 m
Xception	0.5357	2.4954	0.5510	0.5357	0.5371	11.6281	1m19s
VGG16	0.8310	1.4025	0.8477	0.8310	0.8282	13.0543	2m53s
ResNet50	0.9286	0.2400	0.9339	0.9286	0.9285	13.3521	54s
ResNet101	0.9633	0.2061	0.9672	0.9633	0.9626	13.4807	1m18s
Google VIT	0.9857	0.0466	0.0466	0.9865	0.9854	13.5882	9m28s

Table 2 Hyperparameters and fine-tuning of each model input dimensions

Input Dimensions	CNN (256, 256, 3)	Inceptionv3 (256,256,3)	Xception (299, 299, 3)	VGG16 (256, 256, 3)	ResNet50 (256,256,3)	ResNet101 (256,256,3)	Google VIT (256,256,3)
Layers	• Conv2D (64, 128, 256, 512 filters, 3×3 kernel) • Max-Pooling2D (2×2) • Global-Average-Pooling2D • Dense (21)	• Flatten • Batch Normalization (momentum=0.99, epsilon=0.001) • Dense (1024) • Dense (21)	• Dense (128) • Dense (128) • Dense (64) • Dense (32) • Flatten • Dense (21)	• Dense (256) • Dense (64) • Flatten • Dense (16) • Dense (21)	• Dense (128) • Dense (64) • Flatten • Dense (21)	• Dense (256) • Dense (128) • Dense (64) • Flatten • Dense (21)	• ViTEncoder • Conv2d(3, 768, (16×16 kernel)) • Dense (768) • Dense (512) • Dense (256) • Dense (128) • Dense (64) • Flatten • Dense (21)
Activation	ReLU, Softmax						Linear, GeLU
Dropout	0.25	0.2 (twice)	0.2 (twice)	0.5	0.2 (twice)	0.2 (thrice)	0
Number of Parameters	4,685,249 (17.87 MB)	248,952,673 (949.68 MB)	21,930,505 (83.66 MB)	15,355,745 (58.58 MB)	24,657,409 (94.06 MB)	44,613,377 (170.19 MB)	86,567,656 (330.30 MB)
Learning Rate	0.001	0.001	0.001	0.0001	0.001	0.001	0.0001
Optimizer	Adam						
Batch Size	24	10	6	8	24	12	16
Loss Function	sparse_categorical_crossentropy						
ReduceLROnPlateau	<i>monitor:</i> val_loss <i>factor:</i> 0.1 <i>patience:</i> 2 <i>min_lr:</i> 0.000001	<i>monitor:</i> val_loss <i>factor:</i> 0.2 <i>patience:</i> 4 <i>min_lr:</i> 0.000001	<i>monitor:</i> val_loss <i>factor:</i> 0.2 <i>patience:</i> 5 <i>min_lr:</i> 0.00001	<i>monitor:</i> val_loss <i>factor:</i> 0.1 <i>patience:</i> 12 <i>min_lr:</i> 0.000001	<i>monitor:</i> val_loss <i>factor:</i> 0.1 <i>patience:</i> 2 <i>min_lr:</i> 0.00001	<i>monitor:</i> val_loss <i>factor:</i> 0.1 <i>patience:</i> 3 <i>min_lr:</i> 0.00001	-
Learning Rate Reduced	3 times	4 times	2 times	2 times	2 times	2 times	-
Early Stopping patience	<i>monitor:</i> val_loss <i>patience:</i> 6	<i>monitor:</i> val_loss <i>patience:</i> 10	<i>monitor:</i> val_loss <i>patience:</i> 14	-	<i>monitor:</i> val_loss <i>patience:</i> 6	<i>monitor:</i> val_loss <i>patience:</i> 6	<i>monitor:</i> val_loss <i>patience:</i> 5
Number of Epochs	56	152	23	100	24	17	15
Total Time	1m26s	2 m	1m19s	2m53s	54s	1m18s	9m28s
Accuracy	0.7310	0.5476	0.5357	0.8310	0.9286	0.9633	0.9857

To compare the computational efficiency within the CNN models, this study considers the time taken for training and the number of parameters used as a basis for comparison.

- ResNet50 (time: 54s || parameters: 24.6 Mil).
- ResNet101 (time: 1m18s || parameters: 44.6 Mil).
- Xception (time: 1m19s || parameters: 21.9 Mil).
- CNN (time: 1m26s || parameters: 4.6 Mil).
- InceptionV3(time: 2 m || parameters: 248 Mil).
- VGG16 (time: 2 m 53s || parameters: 15.3 Mil).

For quick and efficient deployment, ResNet50 seems to be a viable option with a faster training time and comparatively low computational resources. The higher computational

demands and inferior results (as mentioned before) remove the InceptionV3 and Xception models from comparison to an efficient CNN-based model. Comparing the use of hyperparameter tuning that goes under these models, ResNet models used dropout rates of 0.2 (twice for ResNet50 and thrice for ResNet101) which helped in regularization without losing significant performance. Contributing to better convergence, ResNet models showed effective learning rate reduction strategies (reduced twice). This can be evident by the unstable training dynamics in InceptionV3 and Xception which required more reductions. The confusion matrix of all CNN models is shown in Fig. 10. Comparing the top-performing CNN models (ResNet50 and ResNet101) to the Google ViT, the analysis demonstrates some interesting insights. Google ViT achieved the highest accuracy (98.57%), demonstrating superior capabilities in learning relationships in spatial patterns with the help of the self-attention mechanism of the transformer model. The higher accuracy comes with a trade-off for higher training time (9m28s) and computational resources (86.5 Mil). ResNet50 achieves an accuracy of 92.86% within 54s which provides a balance between performance and resource allocation. The confusion matrix of Google ViT model is shown in Fig. 11. Building upon the previous information, the Google ViT model reported the lowest loss (0.0466) which is significantly low compared to all the CNN models. One thing to note here is that Google ViT comes with higher complexity and training time which could potentially lead to overfitting due to the small dataset of UC Merced Use Dataset. ResNet101 with a loss of 0.2061 and ResNet50 with a loss of 0.2400 signifies a stable convergence pattern as in Table 2.

Processing images as a sequence of patches requires more training time in transformer-based architectures which is also evident by the highest training time (9m28s) of Google ViT in this study along with a high number of parameters (86.5 Mil). Given the higher training time, the Google ViT model compensates for it by performing better in terms of F-1 score (98.54%) and Dice coefficient (13.5882), demonstrating a stronger ability to classify land-use images while preserving spatial consistency.

4.1 Hyperparameter tuning and layer modifications

The following Table 2 summarizes the modifications made to the architectures, as well as the values of the key hyperparameters for each model:

4.2 Comparison metrics

For visual comparisons of the models to best understand how they perceive the images, we have generated saliency maps (using Grad-CAM). The interpretation of the saliency maps is guided by the colour gradient in Fig. 9, where the regions of high model attention are highlighted with red and orange regions indicating areas of maximum importance, while blue regions receiving minimal attention; precisely showing the areas of interest in a randomly picked test image. The models pay the most attention to the red-dish gradients and the blue gradients receive the minimum attention. The saliency plots have been extracted from the final layer of each model before they make the predictions. The areas of most attention in each model are enclosed within red bounding boxes along with the class predictions. The tradition CNN model exhibited a clear performance deficit and failed to generalize effectively on this specific task. ResNet101 emerged as the most promising model as it consistently focused on the most relevant regions of the image during classification. The region of most attention is highlighted in red/orange

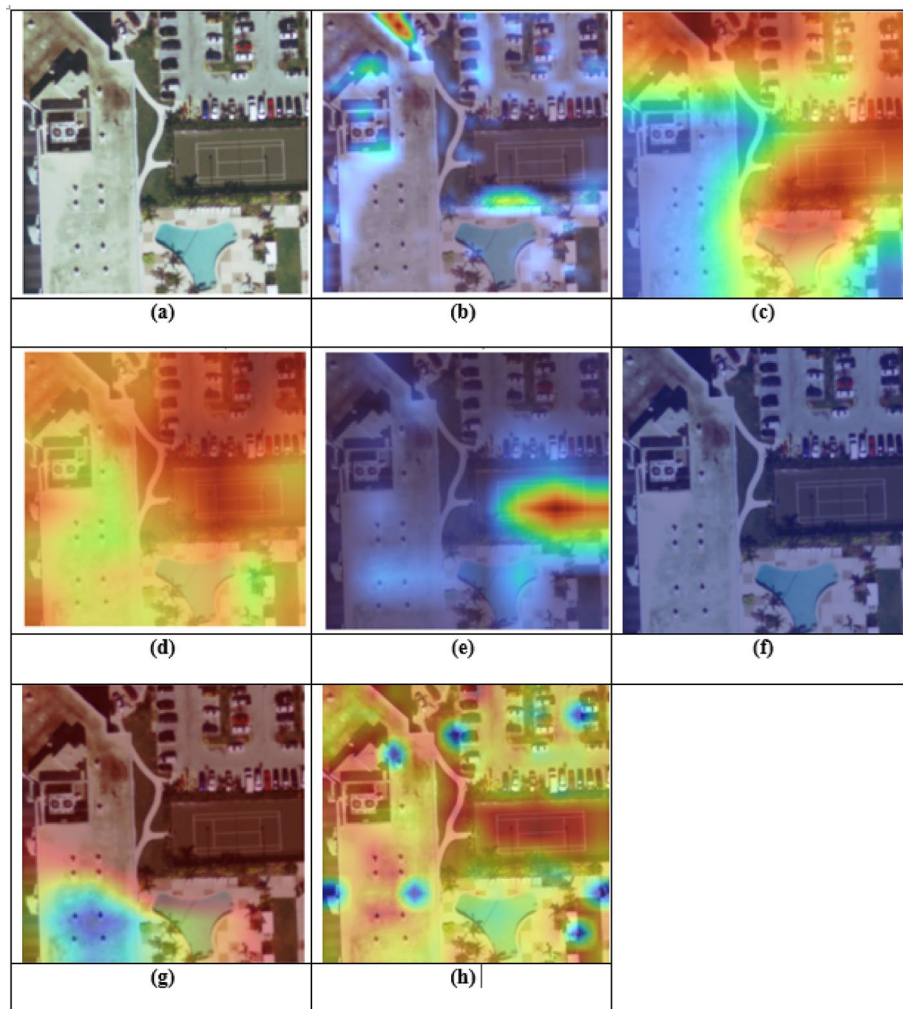


Fig. 9 Saliency map (Grad-CAM) plots on a test image for the performing models. **a** Original Test Image (Class: Tennis Court), **b** CNN (Prediction: Buildings), **c** Inception V3 (Prediction: Tennis Court), **d** ResNet50 (Class: Tennis Court), **e** ResNet101 (Prediction: Tennis Court), **f** VGG16 (Prediction: Tennis Court), **g** Exception (Prediction: Tennis Court), **h** Google ViT (Prediction: Tennis Court).

for ResNet50, ResNet101 and Google ViT. This precisely shows the areas of interest in a randomly picked test image, with the models being the most relied on reddish gradients and lower gradients to blue receiving minimum attention. For comparing the performance across all the models on the image dataset of the UC Merced Use Dataset we considered the following metrics:

- *Hyperparameters* Key hyperparameters such as learning rate, batch size, number of epochs, optimizer type, and dropout rates were used to provide a structured comparison and understand the impact on model performance. Analysis of hyperparameters provides substantial insights into how well these architectures can generalize and still maintain robustness.
- *Time Taken* This metric helps understand the computational efficiency of a model in training and inference which is critical to understand the scalability of such architectures in real-world applications. Live monitoring and classification for large datasets are some of the few examples where efficient inference and training times are crucial for engaging and improving the operational efficiency of the system.

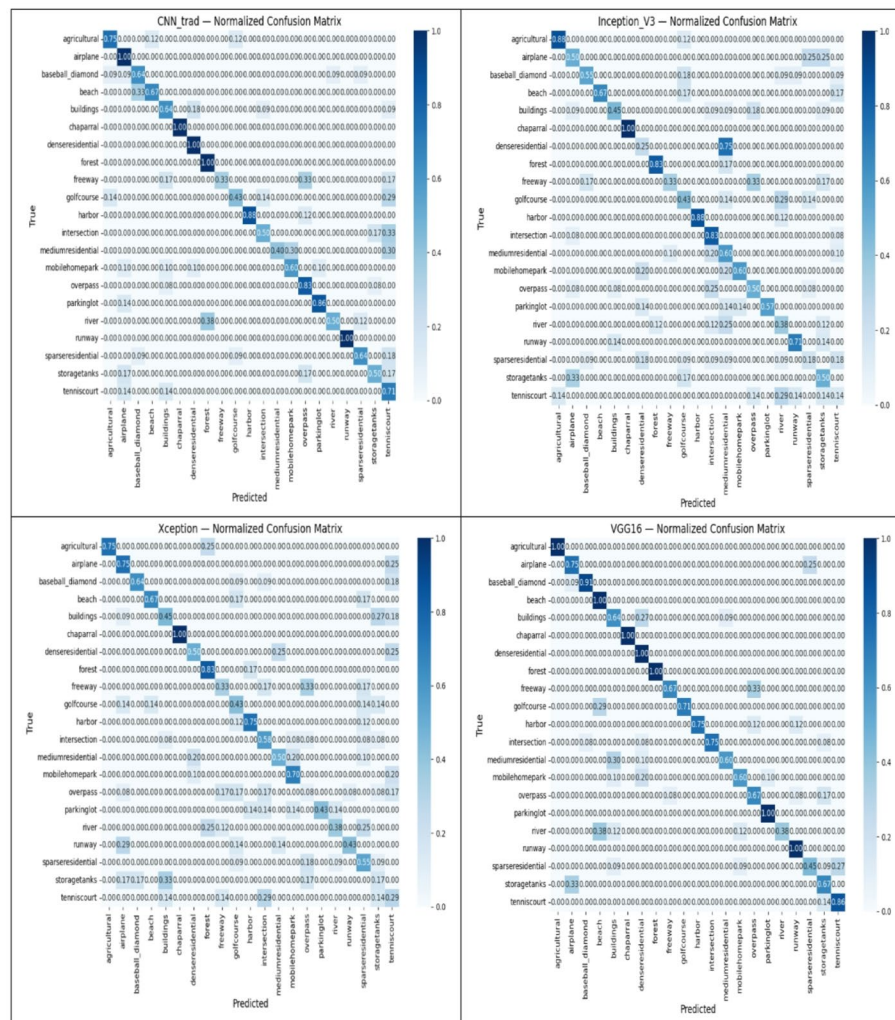


Fig. 10 Confusion matrix of all CNN models

- **Loss** Loss helps this study investigate how distant the predictions of the model are from the actual solution. With regards to the study, this metric is useful in providing details regarding the optimization process of models over variations of hyperparameters, which overall contributes to the assessment being holistic and fair.
- **F1-Score** As a widely known measure, the F1-score provides an overview of the balance in a model's decisions to correctly identify target labels and mitigate false positives and false negatives, which are particularly important in multi-class classification scenarios.
- **Dice Coefficient** This metric is commonly used to evaluate the overlap of ground truth labels with respect to the predicted labels. The assessment of spatial similarity in classified regions of this metric helps maintain spatial accuracy over a diverse range of images.

5 Conclusion and future work

The results of the study showed that the usage of transformers for satellite image classification can be a viable option. However, this study also gave insights into how older and smaller models like ResNet101 can be highly promising. Hence, it is applicable in

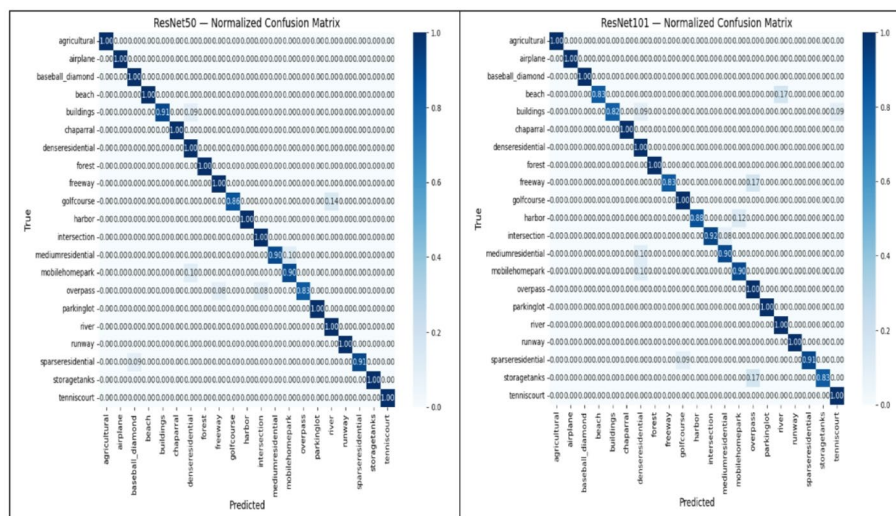


Fig. 10 (continued)

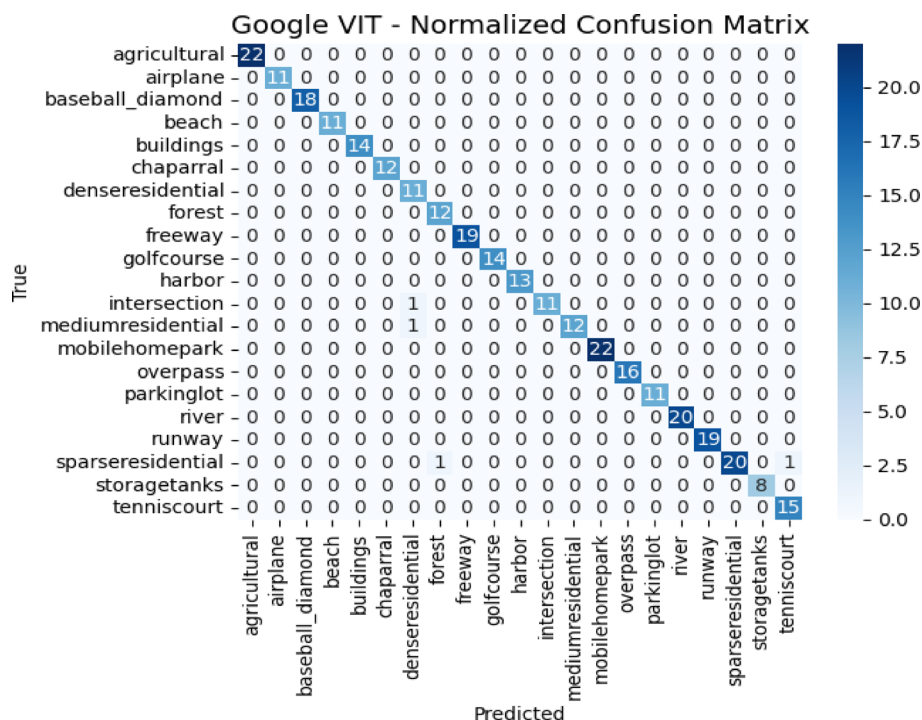


Fig. 11 Confusion matrix of VIT model

situations where the need for faster and lesser computational load is required on the system while sacrificing some robustness as well as options in usability. Results also showed that the ResNet101 emerges as the best-performing model in terms of overall comparisons, providing competitive results with significantly lower training time (1 m 18s). In terms of accuracy, Google VIT achieved the highest number but at a cost of higher computation (86.5 Mil parameters) and training time (9 m 28s). Google VIT demonstrates strong feature extraction capabilities, but it remains sensitive to small datasets which need further optimization for real-world applications. For real-world applications, ResNet50 was most efficient due to faster speed (54s) and lower memory

requirements (24.6 Mil parameters). Google VIT offers superior performance and is suitable for applications with high processing power at the cost of low flexibility and time constraints. With these results and analysis, the choice between models can vary between ResNet101, ResNet50, and Google VIT depending on the application requirement. The following point has been discussed in this study.

- If there is a need for fast inference with limited computational resources, then ResNet50 would be an ideal choice.
- If the system can manage moderate computational power, then ResNet101 offers higher precision in generating classification results.
- If accuracy and deep contextual understanding of the dataset is a priority, then Google VIT offers an optimal solution requiring higher computation.
- If a simple classification problem is there where no other AI-based processing is required in the task after classification, then, the use of the ResNet101 or other similar CNN approaches is viable.

This research was conducted on the labelled datasets. Future work can be based on the integration of hyperspectral and multispectral satellite imagery using multi-model transformer techniques and foundation models.

Author contributions

Diya Nigam, Devesh Loungani and Garveet Jain performed the data collection, designed the model and performed the experiments. Mukund Pratap Singh and Anurag Goswami validate the models, wrote the first draft of the manuscript and supervision. Yadendra Pratap Singh validate the models, wrote the final draft, critical revision of the manuscript and supervision.

Funding

Open access funding provided by Manipal University Jaipur. None. APC charges are covered by Manipal University Jaipur under the Open Access agreement for the corresponding author.

Data availability

The data set used in the experiments can be found on: <http://weegeevision.ucmerced.edu/datasets/landuse.html>.

Code availability

The entire code for this study will be released at github after acceptance.

Declarations

Ethics approval and consent to participate

The current study was performed according to the ethical standards of research and publication. Further, the authors certify that the research presented in this article does not involve any human participants or animals.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 11 July 2025 / Accepted: 27 January 2026

Published online: 23 March 2026

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